Solutions Problem set 4

(1) We have three spin-$\frac{1}{2}$ particles. The total spin can be constructed by adding the first two spins and then the third spin.

We get

$$\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$$

$$\frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} = (1 \oplus 0) \otimes \frac{1}{2}$$

$$= \underbrace{\left(1 \otimes \frac{1}{2}\right)}_{\frac{3}{2} \oplus \frac{1}{2}} \oplus \underbrace{0 \otimes \frac{1}{2}}_{\frac{1}{2}}$$

$$\frac{3}{2} \oplus \frac{1}{2} \oplus \frac{1}{2}$$

$\therefore$ The eigenvalues of $\vec{S}^2$ are $\hbar^2 s(s+1)$

where $s = \frac{3}{2}, \frac{1}{2}$

The degeneracy for $s = \frac{3}{2} \Rightarrow 4$

" $s = \frac{1}{2} \Rightarrow 4$

(2) We have the dipole interaction

$$H = \frac{\gamma_1 \gamma_2}{r^3} \left[\vec{S}_1 \cdot \vec{S}_2 - 3 (\vec{S}_1 \cdot \hat{n}) (\vec{S}_2 \cdot \hat{n}) \right]$$

Now $r = R$, $\hat{n} = \hat{z}$ and hence

$$H = \frac{\gamma_1 \gamma_2}{R^3} \left[\vec{S}_1 \cdot \vec{S}_2 - 3 S_{1z} S_{2z} \right].$$

$$H = \frac{\gamma_1 \gamma_2}{R^3} \left[\frac{1}{2} (\vec{S}^2 - \vec{S}_1^2 - \vec{S}_2^2) - 3 S_{1z} S_{2z} \right].$$

The total angular mom states $|s, m\rangle$ are eigenstates of $\vec{S}^2 - \vec{S}_1^2 - \vec{S}_2^2$ with eigenvalue $\hbar^2 (s(s+1) - \frac{3}{2})$

Now the total angular momentum states are also eigenstates of $S_{1z} S_{2z}$ for this problem. To see this note

$$|s\,m\rangle \qquad |s_1 m_1, s_2 m_2\rangle$$

$$|1\,1\rangle = |\tfrac{1}{2}\tfrac{1}{2}, \tfrac{1}{2}\tfrac{1}{2}\rangle$$

$$|1\,0\rangle = \frac{1}{\sqrt{2}}|\tfrac{1}{2}\tfrac{1}{2}, \tfrac{1}{2}\,\tfrac{-1}{2}\rangle + \frac{1}{\sqrt{2}}|\tfrac{1}{2}\tfrac{-1}{2}, \tfrac{1}{2}\tfrac{1}{2}\rangle$$

$$|1\,{-1}\rangle = |\tfrac{1}{2}\,{-\tfrac{1}{2}}, \tfrac{1}{2}\,{-\tfrac{1}{2}}\rangle$$

$$|0\,0\rangle = \frac{1}{\sqrt{2}}|\tfrac{1}{2}\tfrac{1}{2}, \tfrac{1}{2}\,{-\tfrac{1}{2}}\rangle - \frac{1}{\sqrt{2}}|\tfrac{1}{2}\,{-\tfrac{1}{2}}, \tfrac{1}{2}\tfrac{1}{2}\rangle$$

$$S_{1z} S_{2z} |1\,1\rangle = \frac{\hbar^2}{4} |1,1\rangle$$

$$S_{1z} S_{2z} |1\,0\rangle = -\frac{\hbar^2}{4} |1\,0\rangle$$

$$S_{1z} S_{2z} |1\,{-1}\rangle = \frac{\hbar^2}{4} |1\,{-1}\rangle$$

$$S_{1z} S_{2z} |0\,0\rangle = -\frac{\hbar^2}{4} |0\,0\rangle .$$

$\therefore$ $|s, m\rangle$ are eigenstates of the Hamiltonian.

eigenstate	eigenval
$\|1\,1\rangle$	$\frac{\hbar^2\gamma_1\gamma_2}{R^3}\left[\frac{1}{4} - \frac{3}{4}\right] = -\frac{\hbar^2\gamma_1\gamma_2}{2R^3}$
$\|1\,0\rangle$	$\frac{\hbar^2\gamma_1\gamma_2}{R^3}\left[\frac{1}{4} + \frac{3}{4}\right] = \frac{\hbar^2\gamma_1\gamma_2}{R^3}$
$\|1\,{-1}\rangle$	$-\frac{\hbar^2\gamma_1\gamma_2}{2R^3}$
$\|0\,0\rangle$	$\frac{\hbar^2\gamma_1\gamma_2}{R^3}\left[0 + \frac{3}{4}\right] = \frac{3}{4}\frac{\hbar^2\gamma_1\gamma_2}{R^3}$

(3) When we add $\vec{L}$ and $\vec{S}$, the eigen states of $\vec{J}^2$ and J_z can be written as

$$|j=l+\tfrac{1}{2},m\rangle = \alpha\,|l\,(m-\tfrac{1}{2}),\tfrac{1}{2}\tfrac{1}{2}\rangle + \beta\,|l\,(m+\tfrac{1}{2}),\tfrac{1}{2}\,-\tfrac{1}{2}\rangle$$

$$|j=l-\tfrac{1}{2},m\rangle = \alpha'\,|l\,(m-\tfrac{1}{2}),\tfrac{1}{2}\tfrac{1}{2}\rangle + \beta'\,|l\,(m+\tfrac{1}{2}),\tfrac{1}{2}\,-\tfrac{1}{2}\rangle$$

Since both these vectors are orthonormal we expect

$$\alpha^2+\beta^2=1 \qquad \left(\text{Note: } \alpha,\beta,\alpha',\beta' \text{ can be chosen real}\right)$$

$$\alpha'^2+\beta'^2=1 \qquad \alpha\alpha'+\beta\beta'=0$$

Now $\vec{J}^2 = (\vec{L}+\vec{S})^2 = \vec{L}^2+\vec{S}^2+2L_zS_z+L_-S_++L_+S_-$

where we used the fact that
$\vec{L}\cdot\vec{S} = L_zS_z + L_xS_x + L_yS_y$ and $L_-S_++L_+S_- = 2S_xL_x+2S_yL_y$.

We apply $\vec{J}^2$ on both sides of first eq. we get

$$\vec{J}^2|j=l+\tfrac{1}{2},m\rangle = \hbar^2(l+\tfrac{1}{2})(l+\tfrac{3}{2})\,|j=l+\tfrac{1}{2},m\rangle$$

In order to apply $\vec{J}^2$ on the R.H.S we note that

$$\vec{L}^2|l\,(m-\tfrac{1}{2}),\tfrac{1}{2}\tfrac{1}{2}\rangle = \hbar^2 l(l+1)\,|l\,(m-\tfrac{1}{2}),\tfrac{1}{2}\tfrac{1}{2}\rangle$$

$$\vec{L}^2|l\,(m+\tfrac{1}{2}),\tfrac{1}{2}\,-\tfrac{1}{2}\rangle = \hbar^2 l(l+1)\,|l\,(m+\tfrac{1}{2}),\tfrac{1}{2}\,-\tfrac{1}{2}\rangle$$

$$\vec{S}^2|l\,(m-\tfrac{1}{2}),\tfrac{1}{2}\tfrac{1}{2}\rangle = \tfrac{3}{4}\hbar^2\,|l\,(m-\tfrac{1}{2}),\tfrac{1}{2}\tfrac{1}{2}\rangle$$

$$\vec{S}^2|l\,(m+\tfrac{1}{2})\,\tfrac{1}{2}\tfrac{1}{2}\rangle = \tfrac{3}{4}\hbar^2\,|l\,(m+\tfrac{1}{2}),\tfrac{1}{2}\tfrac{1}{2}\rangle.$$

$$L_zS_z\,|l\,(m-\tfrac{1}{2}),\tfrac{1}{2}\tfrac{1}{2}\rangle = \frac{\hbar^2}{2}(m-\tfrac{1}{2})\,|l\,(m-\tfrac{1}{2}),\tfrac{1}{2}\tfrac{1}{2}\rangle$$

$$L_zS_z\,|l\,(m+\tfrac{1}{2}),\tfrac{1}{2}\tfrac{-1}{2}\rangle = -\frac{\hbar^2}{2}(m+\tfrac{1}{2})\,|l\,(m+\tfrac{1}{2}),\tfrac{1}{2}\,-\tfrac{1}{2}\rangle$$

$$L_- S_+ \,|\, l\,(m-\tfrac{1}{2}), \tfrac{1}{2}\,\tfrac{1}{2}\rangle = 0$$

$$L_+ S_- \,|\, l\,(m-\tfrac{1}{2}), \tfrac{1}{2}\,\tfrac{1}{2}\rangle = \hbar^2 \sqrt{(l-m+\tfrac{1}{2})(l+m+\tfrac{1}{2})}\; |\, l\,(m+\tfrac{1}{2}), \tfrac{1}{2}\,{-\tfrac{1}{2}}\rangle$$

$$L_- S_+ \,|\, l\,(m+\tfrac{1}{2}), \tfrac{1}{2}\,{-\tfrac{1}{2}}\rangle = \hbar^2 \sqrt{(l+m+\tfrac{1}{2})(l-m+\tfrac{1}{2})}\; |\, l\,(m-\tfrac{1}{2}), \tfrac{1}{2}\,\tfrac{1}{2}\rangle$$

$$L_+ S_- \,|\, l\,(m+\tfrac{1}{2}), \tfrac{1}{2}\,{-\tfrac{1}{2}}\rangle = 0$$

Now for convenience let us define

$$|\, l\,(m-\tfrac{1}{2}), \tfrac{1}{2}\,\tfrac{1}{2}\rangle = |1\rangle$$

$$|\, l\,(m+\tfrac{1}{2}), \tfrac{1}{2}\,{-\tfrac{1}{2}}\rangle = |2\rangle$$

Then the first eq. becomes

$$|j = l+\tfrac{1}{2}, m\rangle = \alpha|1\rangle + \beta|2\rangle$$

applying $\vec{J}^2$ on both sides we see that

$$\vec{J}^2 |j = l+\tfrac{1}{2}, m\rangle = \left(\vec{L}^2 + \vec{S}^2 + 2L_zS_z + L_-S_+ + L_+S_-\right)\left(\alpha|1\rangle + \beta|2\rangle\right)$$

$$\Rightarrow \hbar^2 (l+\tfrac{1}{2})(l+\tfrac{3}{2})\left(\alpha|1\rangle + \beta|2\rangle\right) = \left(\hbar^2 l(l+1) + \tfrac{3}{4}\hbar^2\right)\left(\alpha|1\rangle + \beta|2\rangle\right)$$

$$+ \cancel{2}\frac{\hbar^2}{\cancel{2}}(m-\tfrac{1}{2})\,\alpha|1\rangle - \cancel{2}\frac{\hbar^2}{\cancel{2}}(m+\tfrac{1}{2})\,\beta|2\rangle$$

$$+ \hbar^2 \sqrt{(l+m+\tfrac{1}{2})(l-m+\tfrac{1}{2})}\;\beta|1\rangle$$

$$+ \hbar^2 \sqrt{(l-m+\tfrac{1}{2})(l+m+\tfrac{1}{2})}\;\alpha|2\rangle$$

$$\Rightarrow (l+\tfrac{1}{2}-m)\,\alpha|1\rangle + (l+\tfrac{1}{2}+m)\,\beta|2\rangle = \sqrt{(l+m+\tfrac{1}{2})(l-m+\tfrac{1}{2})}\left(\beta|1\rangle + \alpha|2\rangle\right)$$

$$\Rightarrow \frac{\beta}{\alpha} = \left(\frac{l+\frac{1}{2}-m}{l+\frac{1}{2}+m}\right)^{1/2}$$

Let us call $\alpha = A$

$$\beta = \left(\frac{l+\frac{1}{2}-m}{l+\frac{1}{2}+m}\right)^{1/2} A .$$

Now $A^2 + \left(\frac{l+\frac{1}{2}-m}{l+\frac{1}{2}+m}\right) A^2 = 1 \Rightarrow A = \frac{(l+\frac{1}{2}+m)^{1/2}}{(2l+1)^{1/2}}$

$\therefore$

$$\alpha = \frac{(l+\frac{1}{2}+m)^{1/2}}{(2l+1)^{1/2}} \qquad \beta = \frac{(l+\frac{1}{2}-m)^{1/2}}{(2l+1)^{1/2}}$$

$$\beta' = \alpha \quad , \quad \alpha' = -\beta .$$

keeps the sign convention for CG coeff.
(atleast for $l > 0$)!

(4) A tensor operator T_k^q transforms under rotation as given by

$$U[R]\, T_k^q\, U^\dagger[R] = \sum_{q'} D^{(k)}_{q'q}\, T_k^{q'}$$

where $U[R] = \exp\left[-\frac{i}{\hbar}\theta\, \hat{n}\cdot\vec{J}\right]$

$$D^{(k)}_{q'q} = \langle k q' |\, U[R]\, | k q \rangle .$$

For an infinitesimal transformation we have

$$U T_k^q U^\dagger \rightarrow \left(1 - \frac{i}{\hbar}\theta\, \hat{n}\cdot\vec{J}\right) T_k^q \left(1 + \frac{i}{\hbar}\theta\, \hat{n}\cdot\vec{J}\right)$$

$$\simeq T_k^q - \frac{i}{\hbar}\theta\, [\hat{n}\cdot\vec{J}, T_k^q]$$

$$D^{(k)}_{q'q} \longrightarrow \delta_{q'q} - \frac{i}{\hbar}\theta\, \langle k q' | (\hat{n}\cdot\vec{J}) | k q' \rangle .$$

This means that T_k^q must satisfy.

$$[\hat{n}\cdot\vec{J}, T_k^q] = \sum_{q'} \langle kq'|\, \hat{n}\cdot\vec{J}\,|kq\rangle T_k^{q'}$$

$$\Rightarrow \quad [J_i, T_k^q] = \sum_{q'} \langle kq'|J_i|kq\rangle T_k^{q'}$$

which means

$$[J_z, T_k^q] = \hbar q\, T_k^q$$

$$[J_+, T_k^q] = \sum_{q'} \langle kq'|J_+|kq\rangle T_k^{q'}$$

$$= \hbar\sqrt{(k-q)(k+q+1)}\; T_k^{q+1}$$

$$[J_-, T_k^q] = \sum_{q'} \langle kq'|J_-|kq\rangle T_k^{q'}$$

$$= \hbar\sqrt{(k+q)(k-q+1)}\; T_k^{q-1}$$

where we used

$$\langle kq'|J_\pm|kq\rangle = \hbar\sqrt{(k\mp q)(k\pm q+1)}\;\delta_{q',q\pm 1}$$

5 We need to show that $\sum_q (-1)^q S_k^q T_k^{-q}$ is a scalar operator.

(1) For $k=1$ $\quad S_{k=1}^0 = S_z \quad S_{k=1}^1 = -\dfrac{(S_1 + iS_2)}{\sqrt{2}}$

$$S_{k=1}^{-1} = +\frac{(S_1 - iS_2)}{\sqrt{2}}$$

where S_1, S_2, S_3 are Cartesian components and $S_{k=1}^1, S_{k=1}^{-1}, S_{k=1}^0$ are Spherical tensor components.

For $k=1$ we have $\sum_q (-1)^q S_k^q T_k^{-q}$ can be written as

$$S_1^0 T_1^0 - S_1^1 T_1^{-1} - S_1^{-1} T_1^1$$

$$= S_z T_z + \frac{(S_1 + iS_2)}{\sqrt{2}} \frac{(T_1 - iT_2)}{\sqrt{2}} + \frac{(S_1 - iS_2)}{\sqrt{2}} \frac{(T_1 + iT_2)}{\sqrt{2}}$$

$$= S_z T_z + S_1 T_1 + S_2 T_2 = \vec{S} \cdot \vec{T} \quad \text{(as expected!)}$$

(2) General Proof: We need to show

$$U\left(\sum_q (-1)^q S_k^q T_k^{-q}\right) U^\dagger = \sum_q (-1)^q S_k^q T_k^{-q}$$

Left Hand Side:

$$\Rightarrow \sum_{q} (-1)^q \underbrace{U S_k^q U^\dagger}_{\sum_{q'} D^{(k)}_{q'q} S_k^{q'}} \underbrace{U T_k^{-q} U^\dagger}_{\sum_{q''} D^{(k)}_{-q''-q} T_k^{-q''}}$$

see proof

Note:

$$U T_k^q U^\dagger = \sum_{q''} D^{(k)}_{q''q} T_k^{q''}$$

$$U T_k^{-q} U^\dagger = \sum_{q''} D^{(k)}_{q''-q} T_k^{q''}$$

$$= \sum_{q''} D_{-q''-q} T_k^{-q''}$$

Now

$$\Rightarrow \sum_q (-1)^q \sum_{q'} D^{(k)}_{q'q} S_k^{q'} \sum_{q''} D^{(k)}_{-q''-q} T_k^{-q''}$$

$$= \sum_{q'q''} \sum_q \left\{ D^{(k)}_{q'q} \underbrace{D^{(k)}_{-q''-q}}_{} (-1)^q \right\} S_k^{q'} T_k^{-q''}$$

Hint in Shankar

$$D^{*(k)}_{q''q} (-1)^{q''-q}$$

L.H.S

$$\sum_{q'q''} \underbrace{\left(\sum_{q} D^{(k)}_{q'q} \, D^{*(k)}_{q''q} \right)}_{\delta_{q'q''} \ (\text{see proof below})} (-1)^{q''} S_k^{q'} T_k^{-q''}$$

$$= \sum_{q'} (-1)^{q'} S_k^{q'} T_k^{-q'} = R.H.S$$

Let us now prove $\displaystyle\sum_{q} D^{(k)}_{q'q} \, D^{*(k)}_{q''q} = \delta_{q'q''}$

Now

$$D^{(k)}_{q'q} = \langle kq' | U[R] | kq \rangle$$

$$D^{*(k)}_{q'q} = \langle kq | U^{\dagger}[R] | kq' \rangle$$

$$D^{*(k)}_{q''q} = \langle kq | U^{\dagger}[R] | kq'' \rangle$$

$\therefore$

$$\sum_{q} D^{(k)}_{q'q} \, D^{*(k)}_{q''q} = \sum_{q} \langle kq' | U[R] \underbrace{| kq \rangle \langle kq |}_{\text{Identity}} U^{\dagger}[R] | kq'' \rangle$$

$$= \langle kq' | U[R] \, U^{\dagger}[R] | kq'' \rangle$$

$$= \delta_{q'q} .$$